

# SQL & MySQL Assignment: Lab-0

## 1. What is SQL and Its Use?

SQL, or Structured Query Language, is the standard language used to communicate with relational database systems. It allows users to store, manipulate, and retrieve data efficiently. The primary purpose of SQL is to provide a consistent way to manage large sets of structured information.

Its uses include:

1. Data Retrieval: Extracting specific information from large datasets using queries.
2. Data Manipulation: Adding new records, updating existing ones, or deleting unnecessary data.
3. Database Definition: Creating and modifying tables, schemas, and other structures.
4. Access Control: Granting or revoking permissions to ensure data security.
5. Transaction Management: Ensuring data consistency and reliability when multiple operations are performed.

## 2. Describe all the different commands of SQL, mention use of it and syntax.

### 1. Data Definition Language (DDL)

- CREATE: Creates a new database or table.

Syntax: CREATE TABLE table\_name (column1 datatype, column2 datatype);

- ALTER: Modifies an existing database object (adds, deletes, or modifies columns).

Syntax: ALTER TABLE table\_name ADD column\_name datatype;

- DROP: Deletes an entire table, view, or database permanently.

Syntax: DROP TABLE table\_name;

- **TRUNCATE:** Removes all records from a table, but keeps the table structure.

Syntax: `TRUNCATE TABLE table_name;`

- **RENAME:** Changes the name of an existing database object.

Syntax: `ALTER TABLE old_table_name RENAME TO new_table_name;`

## **2. Data Manipulation Language (DML)**

- **INSERT:** Adds new rows of data into a table.

Syntax: `INSERT INTO table_name (col1, col2) VALUES (val1, val2);`

- **UPDATE:** Modifies existing data within a table.

Syntax: `UPDATE table_name SET col1 = val1 WHERE condition;`

- **DELETE:** Removes specific records from a table based on a condition.

Syntax: `DELETE FROM table_name WHERE condition;`

## **3. Data Query Language (DQL)**

- **SELECT:** Retrieves data from one or more tables.

Syntax: `SELECT column1, column2 FROM table_name WHERE condition;`

## **4. Data Control Language (DCL)**

- **GRANT:** Gives permissions to a user to access the database or perform specific tasks.

Syntax: `GRANT SELECT, INSERT ON table_name TO user_name;`

- **REVOKE:** Withdraws the access permissions previously given to a user.

Syntax: REVOKE SELECT, INSERT ON table\_name FROM user\_name;

## 5. Transaction Control Language (TCL)

- COMMIT: Permanently saves all changes made during the current transaction.

Syntax: COMMIT;

- ROLLBACK: Undoes changes made in the current transaction if an error occurs.

Syntax: ROLLBACK;

- SAVEPOINT: Sets a point within a transaction to which you can later roll back.

Syntax: SAVEPOINT savepoint\_name

## 3. Write step by step procedure to install MYSQL.

Step 1: Go to the official MySQL Community Downloads page and select the latest MySQL Installer version for Windows.

The screenshot displays the MySQL website's product offerings. On the left sidebar, there are links for the MySQL Newsletter, Free Webinars (including a recent webinar on HeatWave Lakehouse), and Contact Sales. The main content area features MySQL Enterprise Edition, MySQL Cluster CGE, and a highlighted link at the bottom for MySQL Community (GPL) Downloads, which is circled in yellow with a hand cursor icon.

|                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>MySQL Newsletter</b><br>Subscribe »<br>Archive »<br><br><b>Free Webinars</b><br>Deep Dive into MySQL HeatWave Lakehouse<br>Tuesday, August 08, 2023<br><br>Migrating from On-Premises MySQL to MySQL HeatWave On-Demand<br><br>MySQL Security from Data Protection to Regulation Compliance On-Demand<br><br>More »<br><br><b>Contact Sales</b><br>USA: +1-866-221-0634<br>Canada: +1-866-221-0634 | <b>MySQL Enterprise Edition</b><br>MySQL Enterprise Edition includes the most comprehensive set of advanced features, management tools and technical support for MySQL.<br><br>Learn More »<br>Customer Download »<br>Trial Download »<br><br><b>MySQL Cluster CGE</b><br>MySQL Cluster is a real-time open source transactional database designed for fast, always-on access to data under high throughput conditions. <ul style="list-style-type: none"><li>• MySQL Cluster</li><li>• MySQL Cluster Manager</li><li>• Plus, everything in MySQL Enterprise Edition</li></ul><br>Learn More »<br>Customer Download » (Select Patches & Updates Tab, Product Search)<br>Trial Download »<br><br><a href="#">MySQL Community (GPL) Downloads »</a> |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Step 2: Choose the installer type. You can either download the smaller web installer (which fetches components during setup) or the full installer (which contains all components).

[General Availability \(GA\) Releases](#) [Archives](#) 

## MySQL Installer 8.0.35

 **Note:** MySQL 8.0 is the final series with MySQL Installer. As of MySQL 8.1, use a MySQL product's MSI or Zip archive for installation. MySQL Server 8.1 and higher also bundle MySQL Configurator, a tool that helps configure MySQL Server.

Select Version:

Select Operating System:

|                                                                                                            |        |        |                          |
|------------------------------------------------------------------------------------------------------------|--------|--------|--------------------------|
| <b>Windows (x86, 32-bit), MSI Installer</b><br><small>(mysql-installer-web-community-8.0.35.0.msi)</small> | 8.0.35 | 2.1M   | <a href="#">Download</a> |
| <small>MD5: 214df2ccdf83eb5edc6ca7c115792406   <a href="#">Signature</a></small>                           |        |        |                          |
| <b>Windows (x86, 32-bit), MSI Installer</b><br><small>(mysql-installer-community-8.0.35.0.msi)</small>     | 8.0.35 | 288.6M | <a href="#">Download</a> |
| <small>MD5: 2cfd448a2971b6b5323775ef9e8d012   <a href="#">Signature</a></small>                            |        |        |                          |

Step 3: When prompted, you may log in or sign up for an Oracle account. If you prefer, click on “*No thanks, just start my download*” to continue without an account

## MySQL Community Downloads

Login Now or Sign Up for a free account.

An Oracle Web Account provides you with the following advantages:

- Fast access to MySQL software downloads
- Download technical White Papers and Presentations
- Post messages in the MySQL Discussion Forums
- Report and track bugs in the MySQL bug system

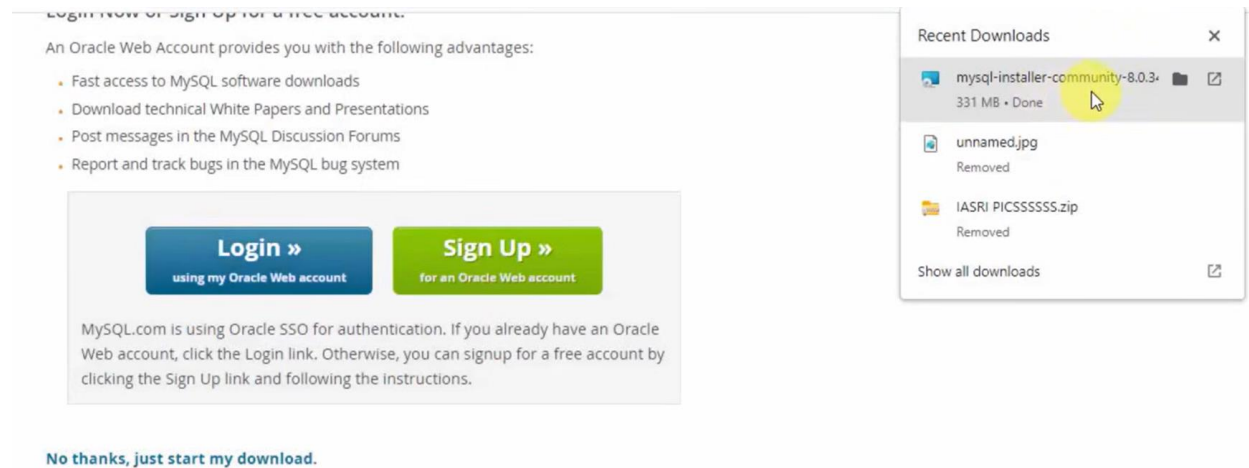
[Login »](#)  
using my Oracle Web account

[Sign Up »](#)  
for an Oracle Web account

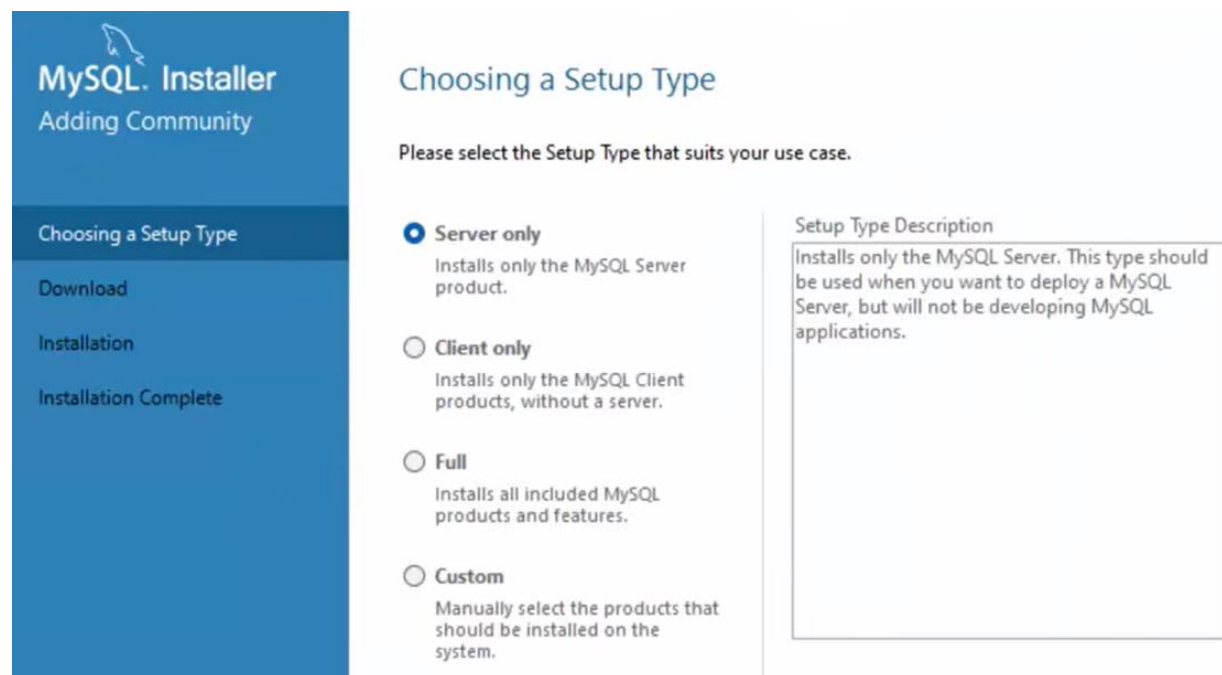
MySQL.com is using Oracle SSO for authentication. If you already have an Oracle Web account, click the Login link. Otherwise, you can signup for a free account by clicking the Sign Up link and following the instructions.

[No thanks, just start my download.](#)

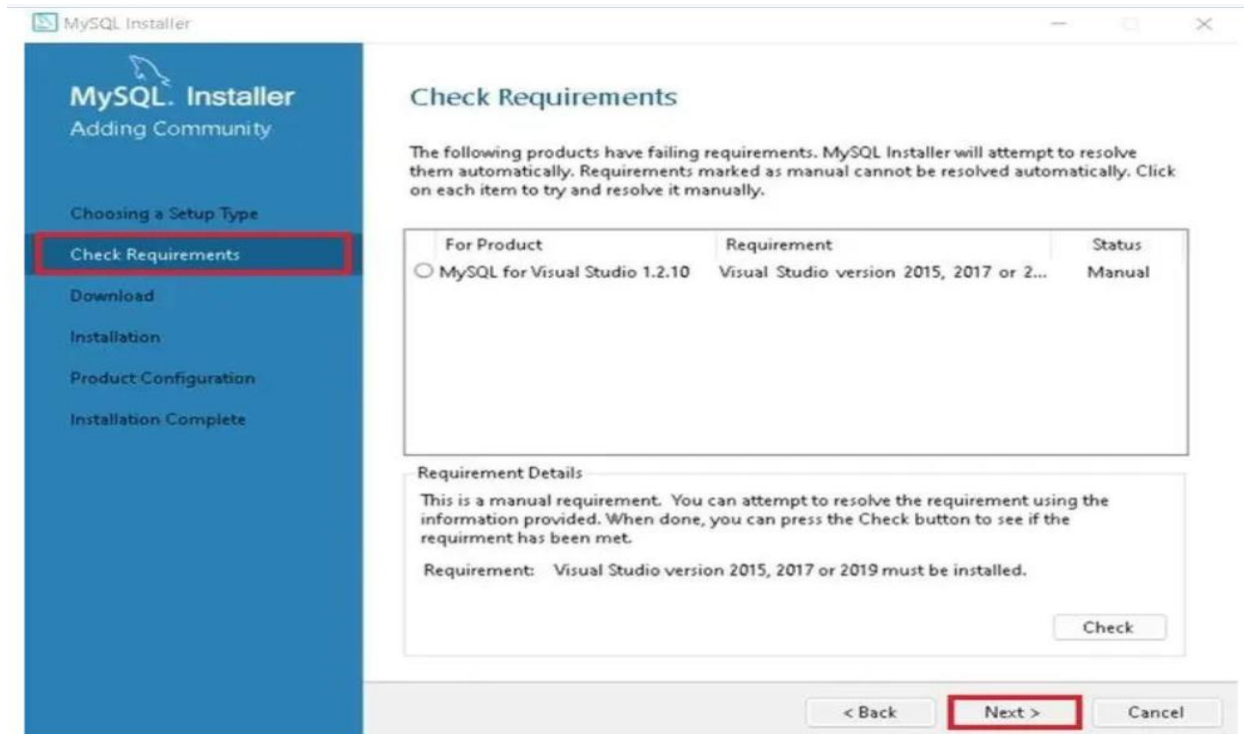
Step 4: Once downloaded, run the **.msi** installer file to launch the MySQL Installer wizard.



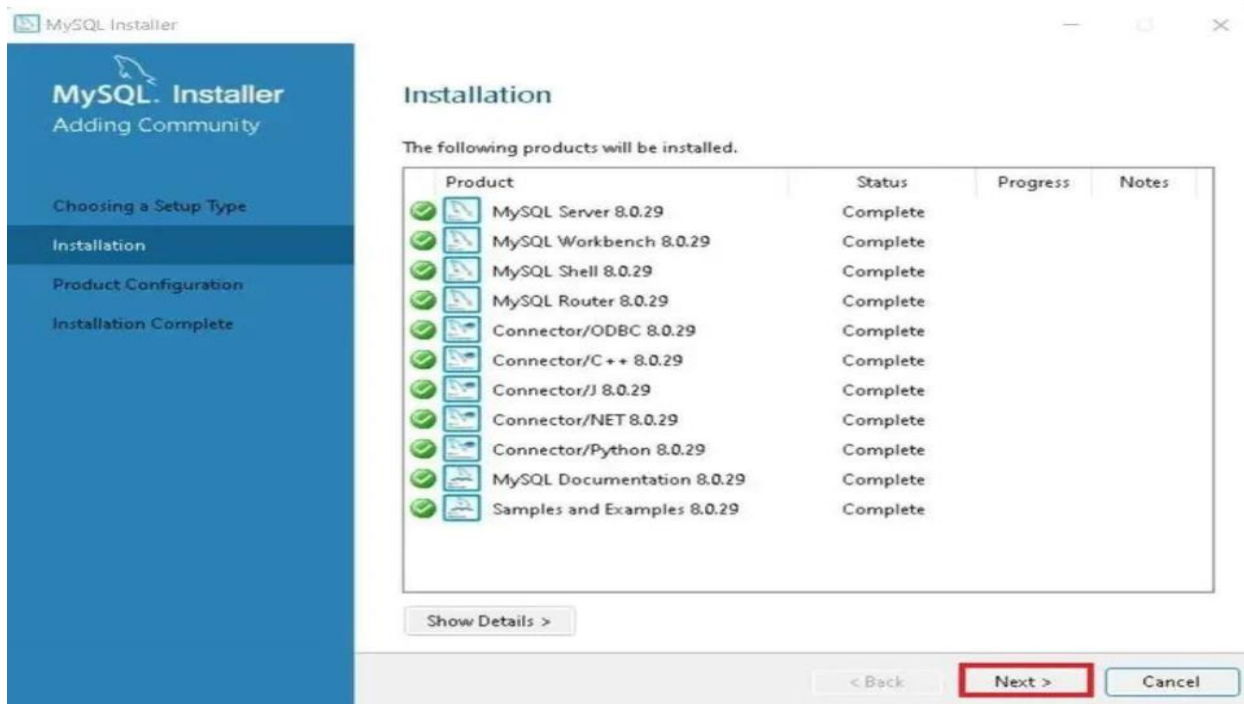
Step 5: Select the setup type. Options include Developer Default, Server Only, Client Only, or Custom. Choose the one that suits your requirement.



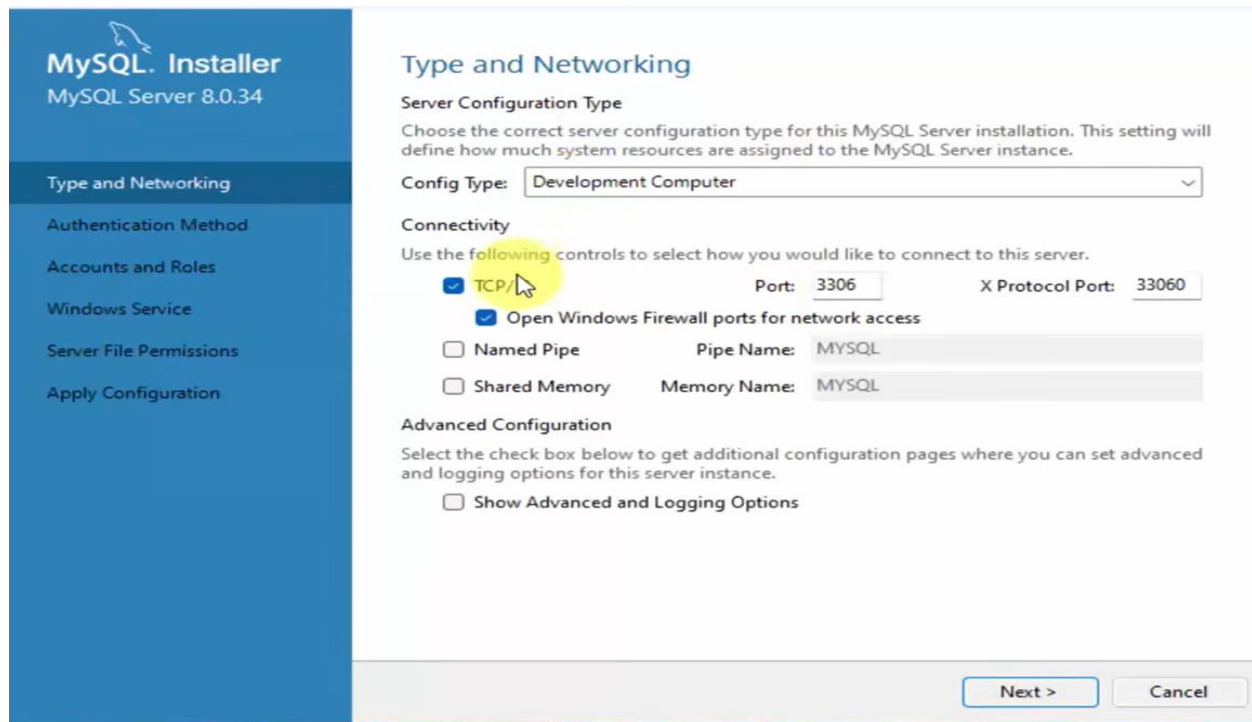
Step 6: The installer will check system requirements. If any prerequisites are missing (such as Visual Studio for MySQL for Visual Studio integration), resolve them before proceeding.



Step 7: The installer will now download and install the selected MySQL products such as MySQL Server, Workbench, Shell, Router, Connectors, Documentation, and Samples.



Step 8: Configure the MySQL Server type and networking. Select whether the server will run as a standalone or clustered server, and confirm the default port (3306).

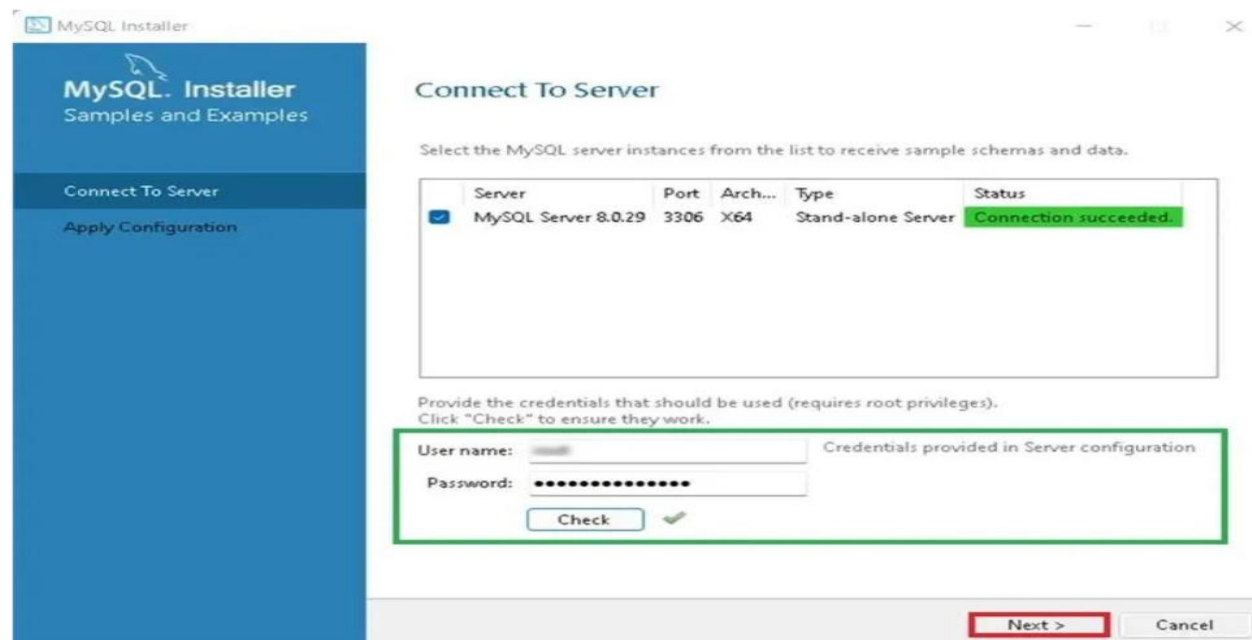


The screenshot shows the 'Type and Networking' configuration window of the MySQL Installer. The left sidebar lists the installation steps: 'Type and Networking' (selected), 'Authentication Method', 'Accounts and Roles', 'Windows Service', 'Server File Permissions', and 'Apply Configuration'. The main area is titled 'Type and Networking' and contains the following sections:

- Server Configuration Type:** A dropdown menu set to 'Development Computer'. Below it, a note states: 'Choose the correct server configuration type for this MySQL Server installation. This setting will define how much system resources are assigned to the MySQL Server instance.'
- Connectivity:** A section with the instruction 'Use the following controls to select how you would like to connect to this server.' It includes:
  - ☒ **TCP/IP**: The 'Port' is set to 3306 and the 'X Protocol Port' is set to 33060.
  - ☒ **Open Windows Firewall ports for network access**
  - ☐ **Named Pipe**: The 'Pipe Name' is set to 'MYSQL'.
  - ☐ **Shared Memory**: The 'Memory Name' is set to 'MYSQL'.
- Advanced Configuration:** A section with the instruction 'Select the check box below to get additional configuration pages where you can set advanced and logging options for this server instance.' It includes:
  - ☐ **Show Advanced and Logging Options**

At the bottom right, there are 'Next >' and 'Cancel' buttons.

Step 9: Choose the authentication method. Strong password encryption is recommended for better security.



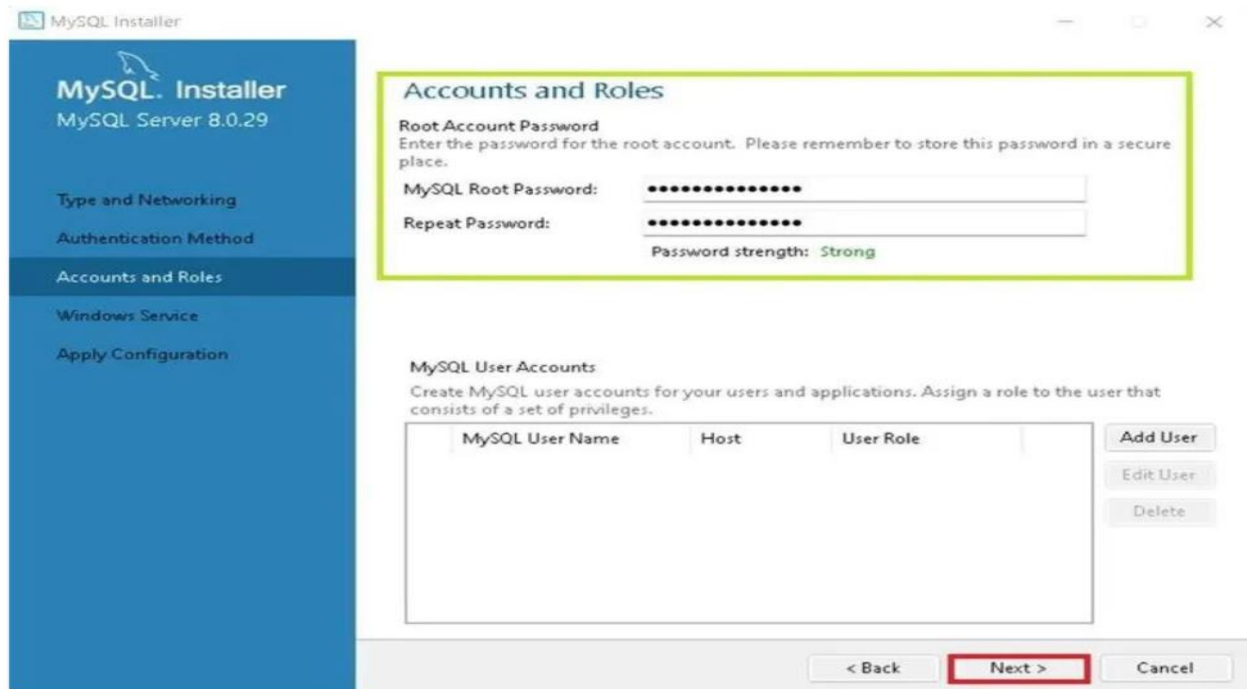
The screenshot shows the 'Connect To Server' configuration window of the MySQL Installer. The left sidebar lists the installation steps: 'Connect To Server' (selected) and 'Apply Configuration'. The main area is titled 'Connect To Server' and contains the following sections:

- Select the MySQL server instances from the list to receive sample schemas and data.** A table with the following data:

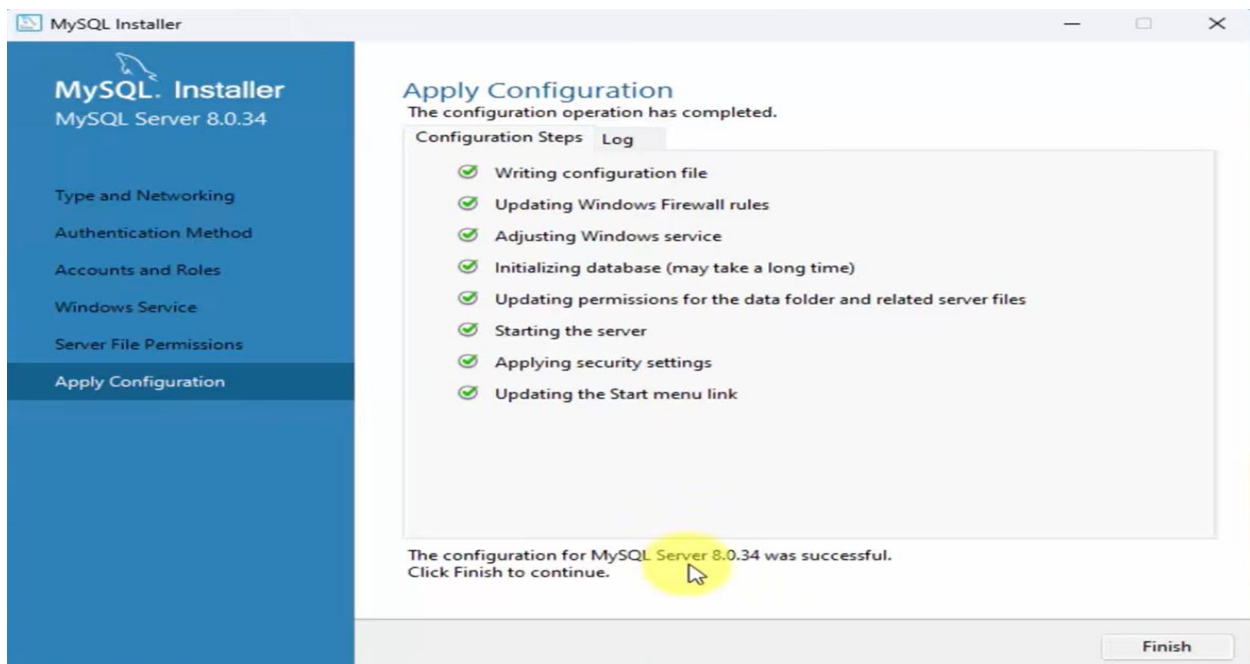
| Server                                                  | Port | Arch... | Type               | Status                |
|---------------------------------------------------------|------|---------|--------------------|-----------------------|
| <input checked="" type="checkbox"/> MySQL Server 8.0.29 | 3306 | X64     | Stand-alone Server | Connection succeeded. |
- Provide the credentials that should be used (requires root privileges). Click "Check" to ensure they work.** A section with a green border containing:
  - User name:** A text field with a dropdown arrow.
  - Password:** A text field with masked characters (dots).
  - A **Check** button and a green checkmark icon.

At the bottom right, there are 'Next >' and 'Cancel' buttons.

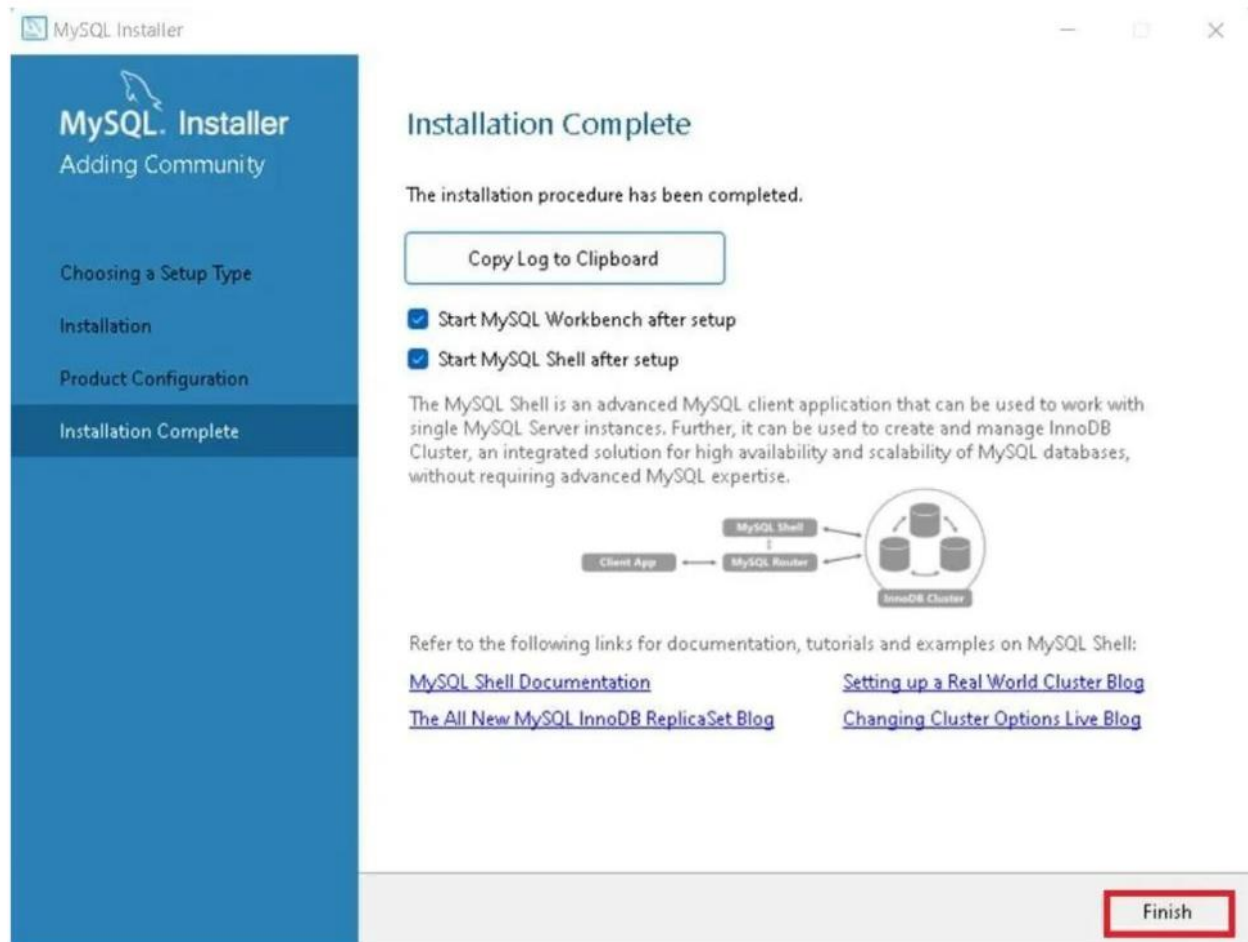
Step 10: Set up the root account by creating a strong password. You may also add additional user accounts and assign roles to them.



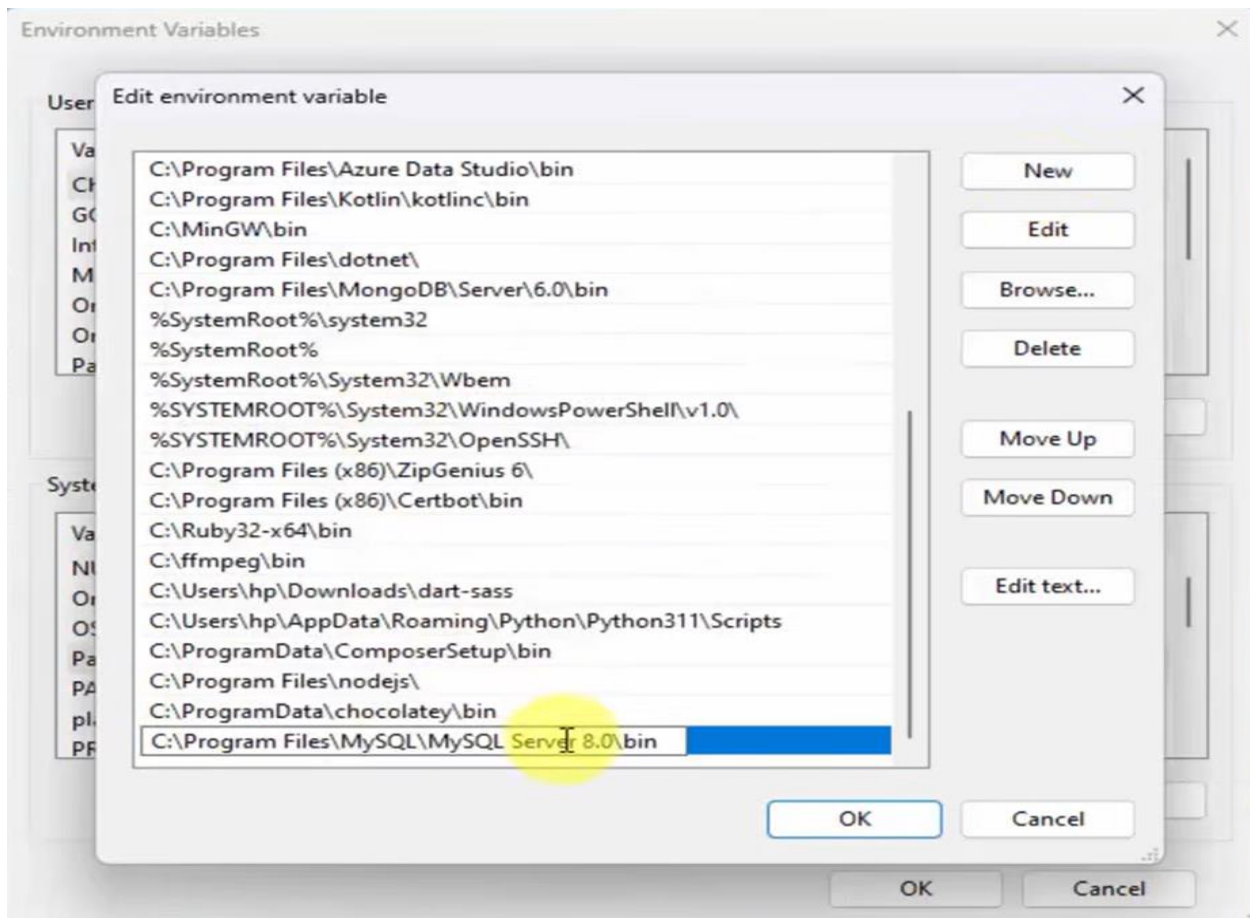
Step 11: Configure MySQL as a Windows Service so that it starts automatically whenever the system boots.



Step 12: Apply the configuration. The installer will test the server connection, finalize the setup, and display a completion screen. You can then launch MySQL Workbench or MySQL Shell to begin using MySQL.



Step 13: Set Environment Variables for MySQL. Add the MySQL bin folder path to the system PATH variable. This allows you to run MySQL commands from Command Prompt without navigating to the installation directory



Step 14: Test MySQL Connection Open Command Prompt, run “mysql -u root -p”, enter your password, and confirm access to the MySQL shell.

```
Command Prompt - mysql -u X + v
Microsoft Windows [Version 10.0.22621.1992]
(c) Microsoft Corporation. All rights reserved.

C:\Users\hp>mysql --version
mysql Ver 8.0.34 for Win64 on x86_64 (MySQL Community Server - GPL)

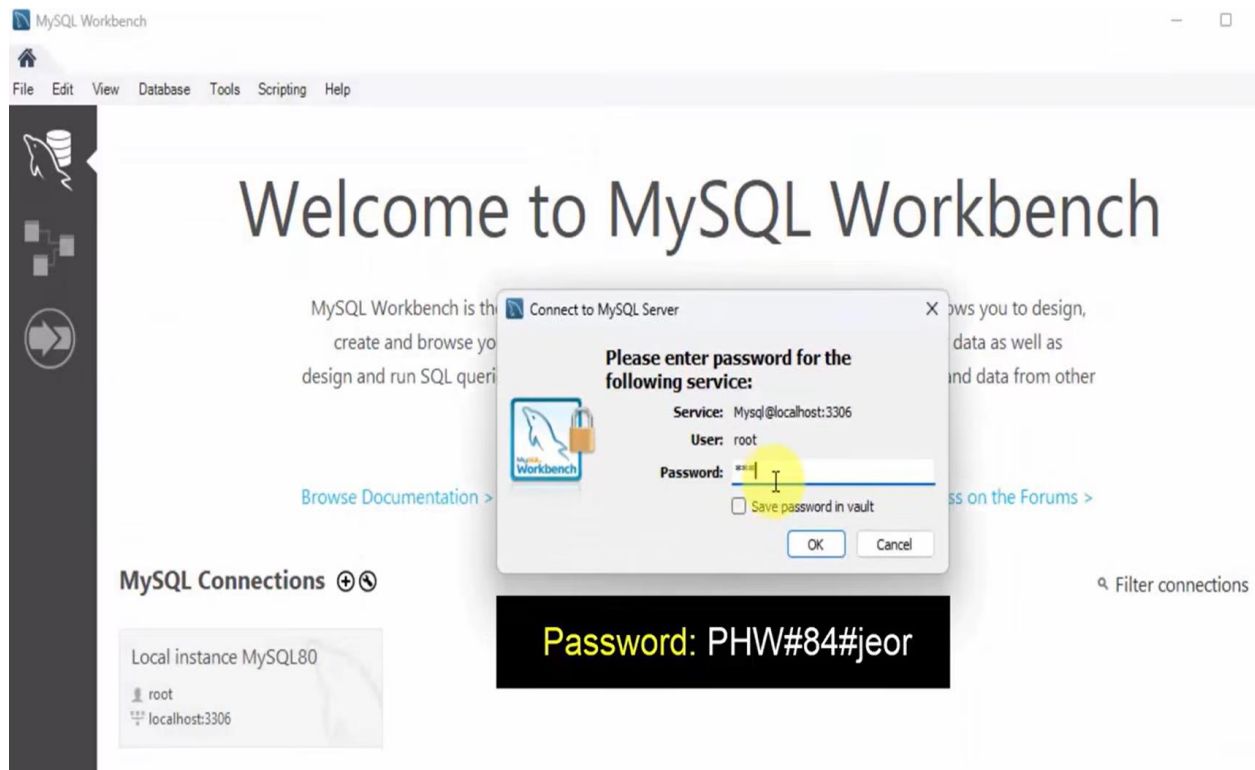
C:\Users\hp>mysql -u root -p
Enter password: *****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 11
Server version: 8.0.34 MySQL Community Server - GPL

Copyright (c) 2000, 2023, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
mysql> |
```

Step 15: Launch MySQL Workbench Open MySQL Workbench from the Start Menu, connect to the local server using the root account, and begin managing databases through the graphical interface.



Step 16: Ready to Use MySQL With installation, configuration, environment setup, and connection testing complete, MySQL is now fully ready to use for creating databases, running queries, and managing data.